

The Dimension of a Vector Space

If a vector space V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V , written as $\dim V$, is the number of vectors in a basis for V . The dimension of the zero vector space $\{\mathbf{0}\}$ is defined to be zero. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.

THEOREM 10

If a vector space V has a basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

THEOREM 11

If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

EXAMPLE 2 Let $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

Find $\dim H$

Solution:

$\dim H$ is the number of vectors in a basis of H . We should find a basis :

- We have only two vectors and we can easily check if they are linearly independent by inspection. $\mathbf{v}_1$ and $\mathbf{v}_2$ are not multiple of each other. Therefore, they are linearly independent (condition 1 is satisfied)
- $\mathbf{v}_1$ and $\mathbf{v}_2$ do span H (given) (condition 2 is satisfied)

Therefore, $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of H . Then, $\dim H = 2$

Alternative Solution:

$\dim H$ is the number of vectors in a basis of H . We should find a basis :

Assume a matrix A whose columns are $\{\mathbf{v}_1, \mathbf{v}_2\}$.

Then find the basis of $\text{Col } A$. those vectors serve as basis for H

We see that both vectors are pivot columns

Therefore, $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of H . Then, $\dim H = 2$

```
A =  
     3     -1  
     6      0  
     2      1  
  
>> rref(A)  
  
ans =  
     1      0  
     0      1  
     0      0
```

EXAMPLE 3 Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \text{ in } \mathbb{R} \right\}$$

$$a \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix} \quad \longrightarrow \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

Construct a matrix A whose columns are the given vectors. Then find the basis of $\text{Col } A$. those vectors serve as basis for H (which is $\text{Col } A$)

$A =$	$\begin{bmatrix} 1 & -3 & 6 & 0 \\ 5 & 0 & 0 & 4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$	$\xrightarrow{\text{rref}(A)}$	$R =$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\xrightarrow{\text{Pivot columns of } A}$	$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$
-------	---	--------------------------------	-------	---	--	--

Therefore, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a basis of H . Then, $\dim H = 3$

Subspaces of a Finite-Dimensional Space

THEOREM 12

Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded, if necessary, to a basis for H . Also, H is finite-dimensional and

$$\dim H \leq \dim V$$

THEOREM 13

The Basis Theorem

Let V be a p -dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V . Any set of exactly p elements that spans V is automatically a basis for V .

The Dimensions of Nul A, Col A, and Row A

The **rank** of a matrix is the dimension of the column space



The number of pivot columns
(number of pivots)

The **nullity** of a matrix is the dimension of the null space



The number of nonpivot columns
(number of free variables)

The **dimension of the row space** is equal to the **rank**

Rank + Nullity =
total number of
columns

EXAMPLE 5 Find the nullity and rank of

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

SOLUTION Row reduce the augmented matrix $[A \ \mathbf{0}]$ to echelon form:

Pivot columns

$$B = \begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three free variables: x_2 , x_4 , and x_5 . Hence the nullity of A is 3. Also, the rank of A is 2 because A has two pivot columns. ■

EXAMPLE 6

- a. If A is a 7×9 matrix with nullity 2, what is the rank of A ?
- b. Could a 6×9 matrix have nullity 2?

SOLUTION

- a. Since A has 9 columns, $(\text{rank } A) + 2 = 9$, and hence $\text{rank } A = 7$.
- b. No. If a 6×9 matrix, call it B , had a two-dimensional null space, it would have to have rank 7, by the Rank Theorem. But the columns of B are vectors in $\mathbb{R}^6$, and so the dimension of $\text{Col } B$ cannot exceed 6; that is, $\text{rank } B$ cannot exceed 6. ■

If nullity is 2, then, the rank must be 7. But a rank of 7 means that we have 7 pivots, which is not possible since there are only 6 rows.

		*	*	*	*	*	*	*	*
0			*	*	*	*	*	*	*
0	0		*	*	*	*	*	*	*
0	0	0		*	*	*	*	*	*
0	0	0	0		*	*	*	*	*
0	0	0	0	0		*	*	*	*

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

m. The columns of A form a basis of $\mathbb{R}^n$.

n. $\text{Col } A = \mathbb{R}^n$

o. $\text{rank } A = n$

p. $\text{nullity } A = 0$

q. $\text{Nul } A = \{\mathbf{0}\}$

Change of Basis

Recall that a coordinate vector is a vector of the weights of a linear combination of the basis to give a certain vector

We can generate the same vector using different basis.
Therefore, the weights of the linear combination will be different for the different basis

THEOREM 15

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases of a vector space V . Then there is a unique $n \times n$ matrix ${}_{\mathcal{C}}P_{\mathcal{B}}$ such that

$$[\mathbf{x}]_{\mathcal{C}} = {}_{\mathcal{C}}P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}} \quad (4)$$

The columns of ${}_{\mathcal{C}}P_{\mathcal{B}}$ are the $\mathcal{C}$ -coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$${}_{\mathcal{C}}P_{\mathcal{B}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix} \quad (5)$$

The matrix ${}_{\mathcal{C}}P_{\mathcal{B}}$ in Theorem 15 is called the **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . Multiplication by ${}_{\mathcal{C}}P_{\mathcal{B}}$ converts $\mathcal{B}$ -coordinates into $\mathcal{C}$ -coordinates.² Figure 2 illustrates the change-of-coordinates equation (4).

Thus $({}_{\mathcal{C}}P_{\mathcal{B}})^{-1}$ is the matrix that converts $\mathcal{C}$ -coordinates into $\mathcal{B}$ -coordinates. That is,

$$({}_{\mathcal{C}}P_{\mathcal{B}})^{-1} = {}_{\mathcal{B}}P_{\mathcal{C}} \quad (6)$$

$$\begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \vdots & \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix} \sim \begin{bmatrix} I & \vdots & {}_{\mathcal{C}}P_{\mathcal{B}} \end{bmatrix}$$

EXAMPLE 3 Let $\mathbf{b}_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\mathbf{c}_1 = \begin{bmatrix} -7 \\ 9 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$, and consider the bases for $\mathbb{R}^2$ given by $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$.

- Find the change-of-coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.
- Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

SOLUTION

- Notice that ${}_{\mathcal{B} \leftarrow \mathcal{C}}P$ is needed rather than ${}_{\mathcal{C} \leftarrow \mathcal{B}}P$, and compute

$$[\mathbf{b}_1 \quad \mathbf{b}_2 \mid \mathbf{c}_1 \quad \mathbf{c}_2] = \begin{bmatrix} 1 & -2 & -7 & -5 \\ -3 & 4 & 9 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & 3 \\ 0 & 1 & 6 & 4 \end{bmatrix} \Rightarrow {}_{\mathcal{B} \leftarrow \mathcal{C}}P = \begin{bmatrix} 5 & 3 \\ 6 & 4 \end{bmatrix}$$

- By part (a) and property (6) (with $\mathcal{B}$ and $\mathcal{C}$ interchanged),

$${}_{\mathcal{C} \leftarrow \mathcal{B}}P = ({}_{\mathcal{B} \leftarrow \mathcal{C}}P)^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -3 \\ -6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -3/2 \\ -3 & 5/2 \end{bmatrix} \quad \blacksquare$$